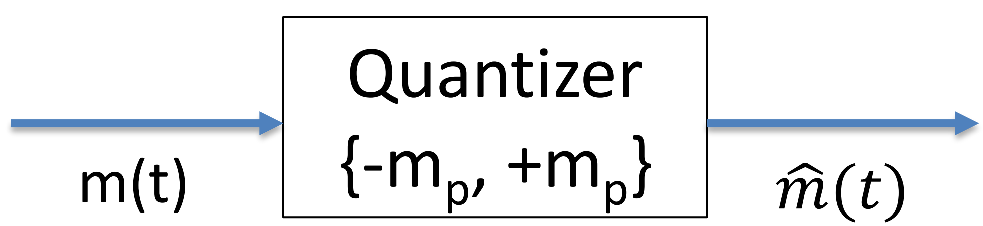
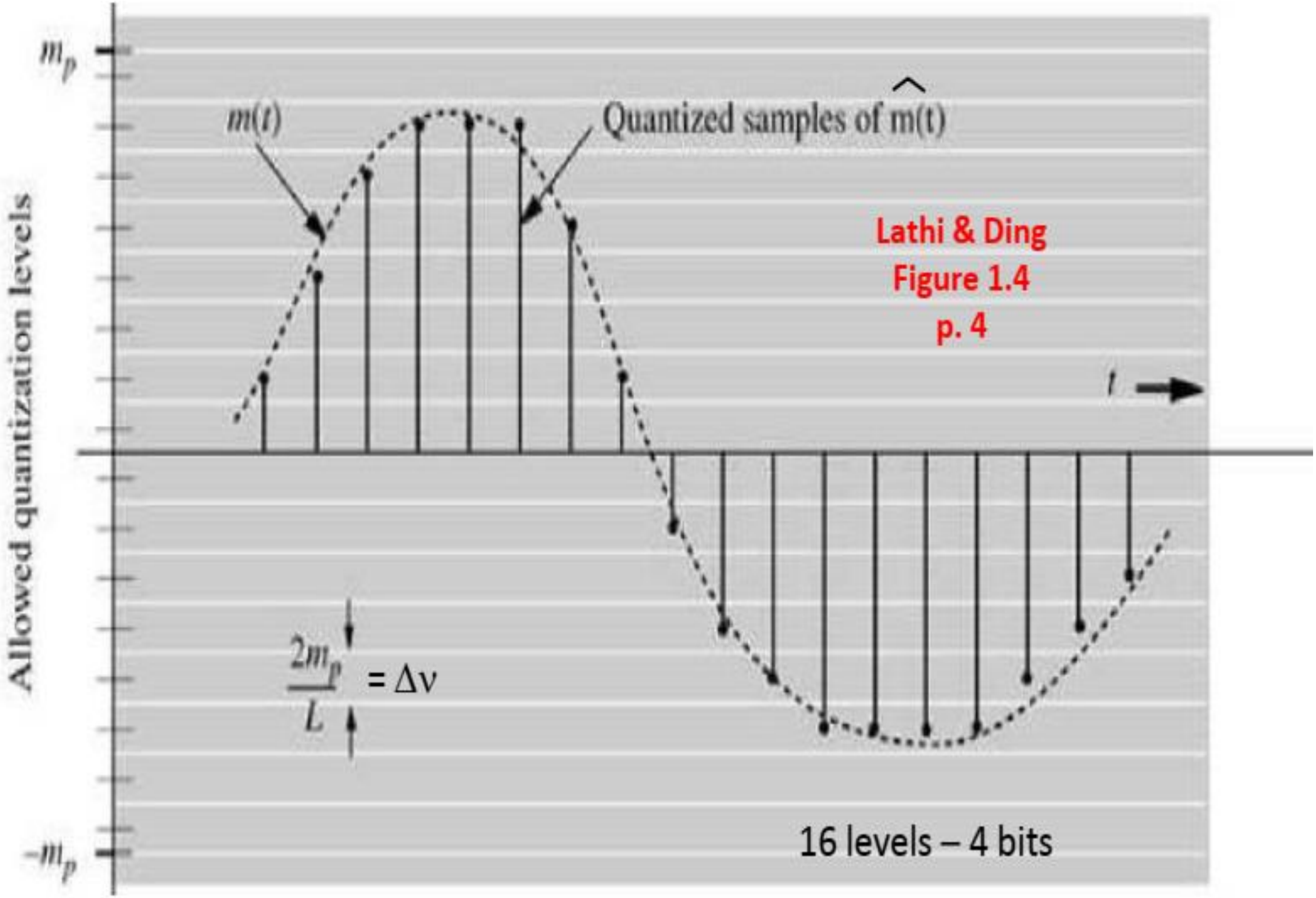
6.2 Quantization

- Quantization is the conversion of the discrete time analog sampled signal to a digital sampled signal whose amplitude values are taken from a finite set of possible amplitudes.
- Quantization may be achieved by rounding off the sample value to one of the closest permissible integer numbers (or quantized levels).
- For quantization, we limit the amplitude of the message signal m(t) to the range $\{-m_p, +m_p\}$.
- m_p is not a parameter of the signal m(t); rather, it is the limit of the quantizer.



- The amplitude range of the quantizer $\{-m_p, m_p\}$ is divided into L uniformly spaced intervals, each of width Δv is thus equal to $\Delta v = 2m_p/L$.
- A sample value is approximated (quantized) by the midpoint of the interval in which it lies. Each sample is now approximated to one of the L numbers (levels). Thus, the signal is converted into digital form.
- The value of the quantized samples taking on any one of the L values. Such a signal is known as an L-ary digital signal.
- The difference between the magnitudes of original and the quantized samples is called the quantization error.
- Quantization causes information loss and this loss is non invertible (i.e., the original signal can't be restored exactly).



m_p
$-m_p$
Allowed quantization levels
$m(t)$
Quantized samples of $\widehat{m(t)}$
Lathi & Ding
Figure 1.4
p. 4
t
$\frac{2m_p}{L} = \Delta v$
16 levels – 4 bits

- The main objective of the quantization process is to minimize the quantization error as much as possible.
- Since a sample value is approximated by the midpoint of the subinterval (of height Δv) in which the sample falls, the maximum quantization error is $\pm\Delta v/2$. Thus, the quantization error lies in the range $(-\Delta v/2, \Delta v/2)$, where

$$\Delta v = \frac{2m_p}{L} \qquad (6.31)$$

- Assuming that the error is equally likely to lie anywhere in the range $(-\Delta v/2, \Delta v/2)$, the mean square quantizing error $\widetilde{q^2}$ is given by:

$$\widetilde{q^2} = \frac{1}{\Delta v}\int_{-\Delta v/2}^{\Delta v/2} q^2\, dq = \frac{(\Delta v)^2}{12} \qquad (6.32)$$

Put (6.31) into (6.32), we get

$$\widetilde{q^2(t)} = \frac{m_p^2}{3L^2} \qquad (6.33)$$

- Because $\widetilde{q^2(t)}$ is the mean square value or the power of the quantization noise, we shall denote it by N_q,

$$N_q = \widetilde{q^2(t)} = \frac{m_p^2}{3L^2}$$

- The quantized signal $\widehat{m}(t)$ at the quantizer output is

$$\hat{m}(t) = m(t) + q(t)$$

where *m(t)* is the input signal to the quantizer, and *q(t) is* the (quantization) noise signal.

- Since the power of the message signal *m(t)* is $\widetilde{m^2(t)}$, then

$$S_o = \widetilde{m^2(t)} \qquad \text{and} \qquad \frac{S_o}{N_o} = 3L^2\frac{\widetilde{m^2(t)}}{m_p^2} \tag{6.34}$$

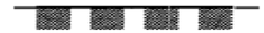
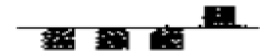
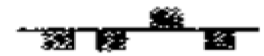
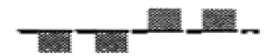
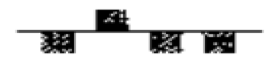
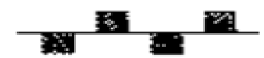
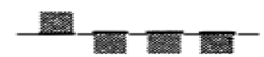
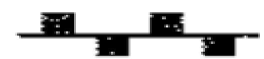
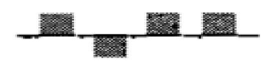
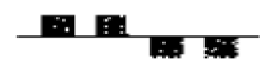
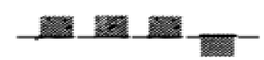
$$N_o = N_q = \frac{m_p^2}{3L^2}$$

S_0/N_0 is the ratio of the signal power to the noise power. It is called the signal to noise ratio (SNR).

- SNR is a very important quantity. It determines the system requirements to achieve a given SNR.
- In Eq. (6.34), m_p is the peak amplitude value that a quantizer can accept, and is therefore a parameter of the quantizer. This means the SNR, is a linear function of the message signal power $\widetilde{m^2(t)}$,

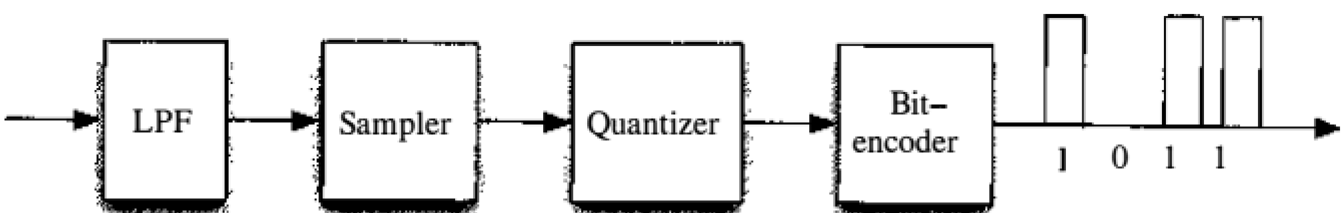
6.3 Coding

- The quantized samples may take L different digital values (L-ary transmission).
- From practical viewpoint, a binary digital signal (a signal that can take on only two values) is very desirable due to its simplicity, economy, and ease of engineering.
- an L-ary signal can be converted into a binary signal by using coding. This is achieved by giving a different binary code word of length n bits to each one of the **L** levels. This is known as the **natural binary code (NBC).**
- $n = \lceil \log_2(L) \rceil$ bits, where $\lceil X \rceil$ is nearest integer $\geq X$.
- For example, if **L** = 16, then, each quantized level can be described uniquely by 4 bits as shown

Digit	Binary equivalent	Pulse code waveform
0	0000	
1	0001	
2	0010	
3	0011	
4	0100	
5	0101	
6	0110	
7	0111	
8	1000	
9	1001	
10	1010	
11	1011	
12	1100	
13	1101	
14	1110	
15	1111	

6.3 PULSE CODE MODULATION (PCM)

- PCM is the most useful and widely used code of all the pulse modulations mentioned.
- PCM basically is a tool for converting an analog signal into a digital signal (A/D conversion).
- It is widely used for transmitting audio (speech) signal digitally. It is also used to store the audio signal to compact disc (CD).
- PCM is performed by three steps: Sampling, Quantization, and Coding.



PCM for audio

- The audio signal bandwidth is about 15 kHz. However, for speech, signal articulation is not affected if all the components above 3400 Hz are suppressed by a low-pass filter.
- The resulting signal is then sampled at a rate of 8000 samples per second. This rate is intentionally kept higher than the Nyquist sampling rate of 6.8 kHz so that realizable filters can be applied for signal reconstruction.
- Each sample is finally quantized into 256 levels ($L = 256$), which requires $\lceil \log_2(256) \rceil = 8$ bits to encode each sample.
- Thus, a digital telephone signal requires $8 \times 8000 = 64{,}000$ binary pulses (bits) per second.

➢ PCM for CD

- Recording the stereo sound on the compact disc (CD) is a more recent application of PCM. The stereo sound consists of two audio sources: one speaker on the left, and one on the right. Each of these is represented by one channel. Each channel has bandwidth 20 kHz.
- Although the Nyquist sampling rate is only 40 kHz, each channel is sampled at a rate of 44.1 kHz in order to use realizable filters as explained before.
- Each channel is then quantized into a large number of quantization levels (L = 65,536) to reduce the quantizing error.
- Each level is encoded by $\lceil \log_2(65{,}536) \rceil = 16$ bits. The encoded stereo sound is finally recorded on the CD at a rate = $44.1 \times 16 \times 2 = 1.4$ Mbit/s.